

Tool Name		Scaphandre	Keple ≥ 0.10	Kepler ≤ v0.9	Green Metric Tool	e-Surgeon (PowerAPI)	BitWatts	PowerJoular	Joular Core	e-Surgeon (elalen)	JoularIX	IgProf (Energy module)	JRAPL	
Granularity	Virtual Machine- Pod-Level	X	X	X	NO	NO	NO	X	X	NO	NO	X	NO	
	Container-Level	X	X	X	X	NO	NO	NO	NO	NO	NO	NO	NO	
	Application-Level	X	X	X	NO	NO	NO	X	X	NO	NO	X	NO	
	Process-Level	X	X	X	NO	X	X	X	X	NO	NO	X	NO	
	Thread-Level	NO	NO	NO	NO	NO	NO	NO	NO	X	X	X	NO	
	Code Section-Level	NO	NO	NO	X(SG functionality)	NO	NO	NO	NO	X	X	X	X	
Measurement Sources	CPU	Hardware-Level Energy Estimation	Hardware-Exposed Energy Interface	External Power Meter	NO	NO	NO	NO	NO	NO	NO	NO	NO	
		Model-based	NO	X(RAPL)	X(RAPL)	X(RAPL)	NO	NO	X(RAPL through powercap)	X(RAPL (linux/windows) or powerMetrics (Mac))	NO	X(RAPL through powercap)	X(RAPL through PAPI)	X(RAPL through linux MSR kernel module)
		Software Activity Share Proxy	NO	XRegression-based models for VM deployment	XRegression-based models for VM deployment	X1) XGBoost model (TDP, Machine Specs and CPUs) 2) Analytical Model(TDP, #CPUchips and CPU %)	X Linear Regression Model (HPC : "instructions", "cache references", "cache misses"	XHPC vector (cycles, instructions, cache misses)	NO	NO	X(TDP) Through PowerAPI	NO	NO	NO
		NO	OS-provided PID CPU % Cgroups Stat	PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing	OS-provided PID CPU % Cgroups Stat Code Special Print (Code Section)	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	1) Time Markers : JVM byte code instrumentation / 2) Call stack	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Call Stack	Call stack	Time Markers	
	DRAM	Hardware-Level Energy Estimation	Hardware-Exposed Energy Interface	External Power Meter	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
		Model-based	NO	X(RAPL)	X(RAPL)	X(RAPL)	NO	NO	X(RAPL)	X(RAPL)	NO	X(RAPL) Through Power Joular	X(RAPL)	X(RAPL through linux MSR kernel module)
		Software Activity Share Proxy	NO	OS-provided PID CPU % Cgroups Stat	PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing	OS-provided PID CPU % Cgroups Stat (Container) Code Special Print (Code Section)	NO	NO	PID CPU %	PID CPU %	NO	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace	Call stack sampling	Time Markers
		NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	
	GPU	Hardware-Level Energy Estimation	Hardware-Exposed Energy Interface	External Power Meter	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
		Model-based	NO	X(NVLM)	X(NVLM)	X(NVLM)	NO	NO	X(NVIDIA-SMI, AMD-SMI) (computers and servers)	X(NVIDIA-SMI, AMD-SMI) (computers and servers)	NO	X(NVIDIA SMI) (computers and servers through Power Joular	NO	NO
		Software Activity Share Proxy	NO	NVLM-provided (I.e. driver) per-process SM utilization	NVLM-provided (I.e. driver) per-process SM utilization	OS-provided PID CPU % Cgroups Stat Code Special Print (Code Section)	NO	NO	PID GPU % (from driver API)	PID GPU % (from driver API)	NO	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace	NO	NO
		NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	
	Network (NIC)	Hardware-Level Energy Estimation	Hardware-Exposed Energy Interface	External Power Meter	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
		Model-based	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
		Software Activity Share Proxy	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
		NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	
	System	Hardware-Level Energy Estimation	Hardware-Exposed Energy Interface	External Power Meter	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
		Model-based	X(RAPL)	NO	NO	X(RAPL PSYS Domain)	NO	NO	NO	X(RAPL)	NO	NO	X(RAPL through PAPI)	X(RAPL through linux MSR kernel module)
		Software Activity Share Proxy	NO	Regression-based model	Regression-based trained power model	X1) XGBoost model (TDP, Machine Specs and CPUs) 2) Analytical Model(TDP, #CPUchips and CPU %)	X(TDP)	X HPC (cycles, instructions, cache misses) vector	polynomial regression For Raspberry Pi	polynomial regression For Raspberry Pi	NO	polynomial regression For Raspberry Pi	NO	NO
		OS-provided PID CPU %	OS-provided PID CPU % Cgroups Stat	PID runtime + HPC vector (cycles, instructions, cache misses) through eBPF tracing	OS-provided PID CPU % Cgroups Stat (container) Code Special Print (Code Section)	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	OS-provided PID CPU %	NO	1) OS-provided PID CPU % for Java Threads through JVM instrumentation 2) Statistical sampling of thread stack trace	Call stack sampling	Time Markers	
Allocation Model	Activity Share Estimation	Directly	X	X	X	X	X	X	X	X (for NIC)	X(for threads)	NO	NO	
		Proxy Reconstruction	NO	NO	X (from HPC vector (cycles, instructions, cache misses))	NO	NO	NO	NO	NO	NO	NO	NO	
		Statistical Sampling	NO	NO	NO	NO	NO	NO	NO	X(for CPU : call-stack sampling version)	X	X(1) Allocate all RAPL system energy to process if it is attached to 2) Statistical Sampling (stack trace) to allocate energy to functions)	NO	
	Execution Time Markers	NO	NO	NO	X(SG functionality)	NO	NO	NO	NO	X (for CPU : bytecode instrumentation version)	NO	NO	X	
Virtualised environment support	Baseline Modelling and Calibration		Modeled and allocated proportionnaly to processes size	Modeled and allocated proportionnaly to processes size	Modeled and allocated to container according to OS-provided PID CPU %	NO	NO	NO	Modeled and subtracted before allocation	NO	NO	NO	NO	
	Pretrained Model		Regression model (# CPU and OS-provided Resource Utilisation metrics as input)	Regression model (# CPU and OS-provided Resource Utilisation metrics as input)	1) XGBoost model (TDP, Machine Specs and CPUs) 2) Analytical Model(TDP, #CPUchips and CPU %)	NO	NO	NO	NO	NO	NO	NO	NO	
	Two instances : host exposing VM-level metrics to guest through shared file		X	NO ((CNCF 2023) mentions a planned "passthrough" mode for VM deployment which expose VM energy in a shared file.	NO ((CNCF 2023) mentions a planned "passthrough" mode for VM deployment which expose VM energy in a shared file.	NO	NO	X	X	NO	NO	NO	NO	
	Delegated to another		NO	NO	NO	NO	NO	NO	NO	NO	Uses PowerJoular	NO	NO	